

DGSNA: prompt-based Dynamic Generative Scene-based Noise Addition method

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Abstract—To ensure the reliable operation of speech systems across diverse environments, noise addition methods have emerged as the prevailing solution. However, existing methods offer limited coverage of real-world noisy scenes and depend on pre-existing scene-based information and noise. This paper presents prompt-based Dynamic Generative Scene-based Noise Addition (DGSNA), a novel noise addition methodology that integrates Dynamic Generation of Scene-based Information (DGSNI) with Scene-based Noise Addition for Speech (SNAS). This integration facilitates automated scene-based noise addition by transforming clean speech into various noise environments, thereby providing a more comprehensive and realistic simulation of diverse noise conditions. Experimental results demonstrate that DGSNA significantly enhances the robustness of speech recognition and keyword spotting models across various noise conditions, achieving a relative improvement of up to 11.21%. Furthermore, DGSNA can be effectively integrated with other noise addition methods to enhance performance. Our implementation and demonstrations are available at <https://dgsna.github.io>.

Index Terms—noise addition method, scene-based noise, scene-based information.

I. INTRODUCTION

Synthesizing noisy speech datasets via noise addition is crucial for enhancing performance on speech-centric tasks such as Automatic Speech Recognition (ASR) and Keyword Spotting (KWS). Constructing scene-based noise [1], crucial for replicating real-world acoustic environments, significantly improves the efficacy of scene-based noise addition methods. This, in turn, enhances a more diverse and realistic simulation of noisy conditions, which can significantly enhance the robustness of downstream speech processing models, such as ASR and KWS, especially in challenging auditory scenarios like the cocktail party effect [2], [3], highlighting its importance in advancing speech signal processing. Scene-based noise addition methods primarily serve two purposes: (a) Enhancing Model Robustness: Real-world speech signals are often corrupted by diverse environmental noises, such as street and office noise. Integrating these noises into training data enables models to better adapt to such conditions, improving

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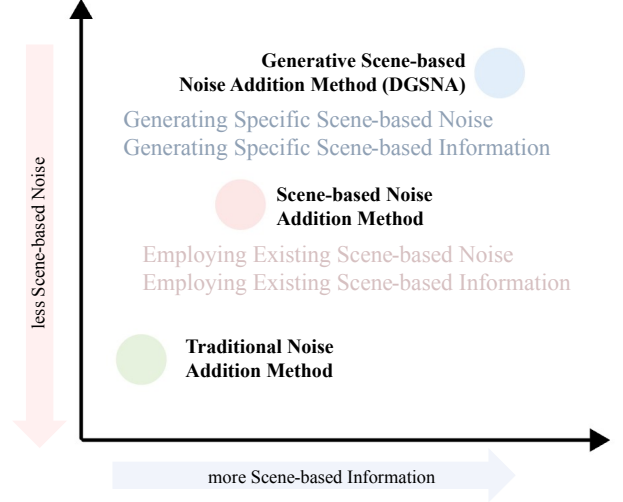


Fig. 1: Comparative analysis of noise addition methods.

robustness. (b) Evaluating Model Performance: Adding scene-based noise allows for performance assessment across various acoustic environments.

Fig. 1. depicts a stage-by-stage comparison of diverse noise addition methods. *Scene-based Information* and *Scene-based Noise* are essential for reproducing a noisy scene. *Scene-based Information* must encompass the scene dimensions, microphone placement, noise source type and location, and human voice source location. *Scene-based Noise* must correspond to the noise source types specified in the *Scene-based Information* and can be further diversified across a broad spectrum of acoustic levels.

Traditional noise addition for speech, which includes generating random noise, simulating with filters, and modifying spectral properties, is effective for data augmentation but cannot incorporate specific environmental noises, limiting its utility in targeted acoustic environments. **Scene-based noise addition** methods, which are categorized into physically constructing scenes and recording the resulting sounds or using Room Impulse Response (RIR) filters to simulate acoustic environments, have made significant progress [1], [4]. RIR filter simulations are increasingly favored due to the time and resource demands of physical scene construction [5], [6]. However, these non-generative scene-based noise addition methods rely on noise recordings from real environments or simulations based on existing scene-based information. While effective in specific, known, or predictable environments, their generalizability and applicability to diverse or unforeseen scenarios remain limited. The vast diversity of real-world

scenes surpasses practical enumeration and precise definition, posing significant challenges for non-generative methods in capturing the full spectrum of acoustic scenarios. Furthermore, replicating acoustic environments is time- and labor-intensive, challenging researchers to collect sufficient scene-based speech data for effective model training.

To address the aforementioned challenges, we propose prompt-based Dynamic **Generative Scene-based Noise Addition** (DGSNA), integrating Dynamic Generation of Scene-based Information (DGSNI) with Scene-based Noise Addition for Speech (SNAS). DGSNA begins by incorporating a textual description of the target scene into a few-shot prompt [7] within the BET framework. Subsequently, a generative chat model [8] processes this description to dynamically generate scene-based information. The identified noise types from the scene-based information are then passed to the Text-To-Audio (TTA) system [9], which generates corresponding scene-based noise. Finally, this noise is further amalgamated with speech and scene-based information, and processed through RIR filters to produce the final scene-based speech output.

A series of experiments were conducted to validate the effects of scene-based information and noise generation, as well as to demonstrate DGSNA's generalizability across various generative dialogue models. DGSNA was integrated into the data preprocessing pipeline, with models trained on clean datasets and evaluated on noisy datasets to assess its effectiveness for ASR and KWS tasks. Results indicated that a 20% Add Noise Rate (ANR) yielded optimal model robustness. Furthermore, DGSNA was compared with other noise addition methods, revealing that DGSNA can be effectively integrated with other methods for enhanced performance. Finally, an analysis of correctly and incorrectly classified samples was performed to detail the process of generating scenario-based speech with DGSNA.

In summary, this paper presents DGSNA, a novel approach aimed at addressing the critical challenge of comprehensively defining and replicating acoustic environments for scene-based noise addition. Our contributions are both innovative and multifaceted, driving significant advancements in the field of intelligent speech processing:

- We present DGSNA, a novel methodology developed to address the limitations of existing scene-based noise addition methods and to enhance the complexity and utility of speech processing systems.
- By leveraging the BET prompt framework and a generative chat model for **scene-based information**, DGSNA.DGSNI effectively addresses the challenges of scene enumeration and description, enabling more accurate and dynamic scene simulation.
- Utilizing the TTA system and RIR filters for **scene-based noise**, DGSNA.SNAS streamlines the laborious and time-consuming process of replicating diverse acoustic environments.
- We rigorously tested and validated DGSNA through a series of experiments focusing on three research areas: a comparative analysis with other noise addition methods, an evaluation of DGSNA's impact on ASR and KWS

performance with varying ANR, and an assessment of DGSNA's generalizability across diverse generative chat models. Furthermore, these conclusions are substantiated through detailed example analyses, demonstrating the validity and reliability of our approach.

The rest of the paper is organized as follows: In Section II, we introduce the related work to set the stage for this study. Our proposed method and framework are presented in Section III. The experimental settings and experimental results are shown in Section IV and Section V, respectively. Section VI concludes this paper.

II. RELATED WORK

A. Traditional noise addition methods

Traditional noise addition methods mainly include random noise generation, filter simulation, and spectral transformation. Random noise generation creates random values following specific distributions (e.g., Gaussian [10], uniform [11], and those generated by Linear Feedback Shift Register (LFSR) algorithms [12]), to mimic noise signal amplitude and frequency features. Filter simulation uses digital filters to reproduce specific noise types, including white [13], pink [14], and brown noise [15], and also permits customizing filter frequency responses to achieve desired noise characteristics. Spectral transformation utilizes Fourier transforms [16] to add specific frequency noise to the original speech signal in the frequency domain. This method also converts elements of the original speech spectrum into noise with minimal change, employing methods like spectral blur [17] and SpecAugment [18]. However, these conventional methods primarily focus on speech data augmentation rather than optimizing performance in specific acoustic environments, which is a significant limitation. To overcome this limitation, researchers have developed scene-based noise addition methods.

B. Scene-based noise addition methods

Scene-based noise addition methods are broadly divided into two approaches: physical scene construction (constructing and recording sounds in real-world scenes) and RIR filter simulation (using RIR filter simulations to reproduce these environments). The latter is increasingly preferred due to the prohibitive time and resources required for physical scene creation. However, simulating effective RIR filters presents challenges. Image Source Method (ISM) [19], calculates sound propagation by mirroring virtual *image sources* across room boundaries. ISM typically assumes an empty, parallelepiped or rectangular room with a fixed absorption rate for all boundaries. However, this simplified approach may not accurately capture the complex acoustics of real-world environments, where reflections are influenced by furniture arrangements and material compositions. Numerous sophisticated methods have been developed to enhance RIR filter quality and realism. Ray-tracing strategies have been explored for computing sound paths using explicit room models [20], [21], while diffuse-based methods approximate late reverberation [22], [23]. Furthermore, neural networks [24], notably Generative Adversarial Networks (GANs) [25], have been employed to

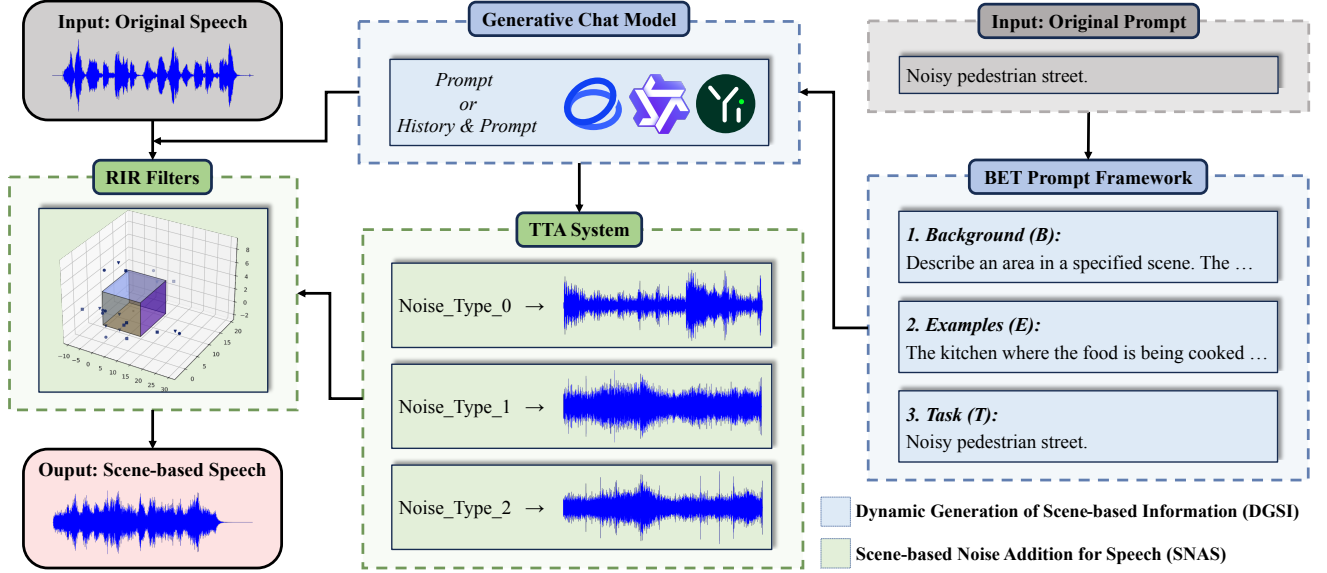


Fig. 2: Overall framework of the proposed DGSNA method. In this framework, the *BET Prompt Framework* module is structured into two forms (for details, see Section V-C). Additionally, the *Generative Chat Model* module incorporates a design pattern that integrates the logos of three different generative chat models, enhancing adaptability and functionality (for details, see Section IV-C).

refine RIR filter simulation, aiming to better approximate the distribution of real recorded RIR filters [26].

Current research on scene-based noise addition largely focuses on known or predictable scenes, often referred to as non-generative scene-based noise addition. A principal limitation of these methods is their inability to capture the extensive diversity of real-world scenes, which exceeds practical cataloging and description capabilities. Moreover, the creation or simulation of known scenes and recording scene-based noise are both labor- and time-intensive processes. This complexity constrains researchers' efforts to collect sufficient scene-based speech data for effective model training. Inspired by the success of generative chat models across text [27], audio [28], and image modalities [29], this paper introduces DGSNA, a novel generative scene-based noise addition method.

C. Generative scene-based noise addition methods

Generative scene-based noise addition methods rely on large language models (LLMs) [30]–[32] fine-tuned via alignment methods such as Reinforcement Learning from Human Feedback (RLHF) and Supervised Fine-Tuning (SFT) [33], [34]. These models, foundational to modern AI, excel in complex reasoning, problem-solving, and creative tasks via natural language interfaces, enabling functions like dialogue, question answering, and artistic content generation [35], [36]. They can engage in natural language conversations, answer questions, provide information, and even generate artistic content like stories, poems, and music. Building on this foundation, this paper proposes DGSi, which uses prompts to guide generative chat models in producing specific scene-based information.

After generating the scene-based information, audio generative model can produce scene-based noise that aligns with these scene-based information specifications. TTA is an emerging application focused on synthesizing diverse audio samples

from text prompts. Early TTA models, as noted in [37], [38], primarily used one-hot labels, which constrained the label space and generative capabilities, leading to monotonous audio outputs. Conversely, natural descriptive text provides richer and more detailed information for audio synthesis. Recent advances in diffusion-based generative models [39], [40] have demonstrated remarkable content creation and comprehension capabilities. The first diffusion-based TTA system, DiffSound [9], significantly outperformed previous models by generating quantized discrete tokens from mel-spectrograms via diffusion. AudioGen [41], using an autoregressive model in discrete waveform space, later surpassed DiffSound. Subsequently, AudioLDM [42], inspired by [43], pioneered continuous Latent Diffusion Models (LDMs), achieving superior quality and efficiency compared to discrete token-based systems. Leading TTA systems like Tango [44] and Make-an-Audio [45] have since integrated LDMs, improving latent space denoising. Building on these advancements, this paper proposes SNAS, which integrates TTA-generated scene-based noise into the speech signal to create scene-based speech.

III. METHODOLOGY

A comprehensive overview of the DGSNA is provided in Fig. 2. In the DGSi component, the text description of the target scene is integrated into the BET prompt framework. This process generates a BET prompt that is subsequently fed into generative chat models to dynamically generate **scene-based information**. Within the SNAS component, the process is initiated by generating **scene-based noise** through TTA systems based on the types of scene-based noise identified in the scene-based information. Subsequently, RIR filters are used to combine this scene-based noise with the scene-based information and the original speech, thus providing comprehensive scene-based speech.

Example of BET prompt for BlueLM-Chat	
Prompt = "	
[[Human]]:	Describe an area in a specified scene. The response must include the dimensions of the area, microphone location, voice location, and noise source location. Each location must conform to normal logic and not exceed the dimensional range. At the same time, there must be at least two types of noise sources. The kitchen where the food is being cooked.
[[AI]]:	An area in the kitchen [kitchen] with (4, 2.5, 4). The microphone at (3.5, 0.5, 1.2), the voice at (2, 1.5, 1.6), the sound of running water [the sound of running water] at (0.5, 0.5, 1.2), the sound of cutting vegetables [the sound of cutting vegetables] at (0.5, 0.5, 1.2), the sound of range hood [the sound of range hood] at (2, 0.5, 2).
[[Human]]:	The living room where a group of people are chatting.
[[AI]]:	An area in the living room [living room] with (8, 5, 4). The microphone at (4, 2.5, 1.2), the voice at (1, 0.75, 1.2), the sound of speaking [the sound of speaking] at (0.75, 2.5, 1.2), the sound of TV [the sound of TV] at (4, 4.5, 1.2).
[[Human]]:	The bathroom where someone is taking a shower.
[[AI]]:	An area in the bathroom [bathroom] with (5, 3, 4). The microphone at (0.5, 0.5, 1.2), the voice at (4, 1.5, 1.6), the sound of shower [the sound of shower] at (4, 2.5, 2), the sound of music [the sound of music] at (0.5, 1, 1.2).
[[Human]]:	Noisy balcony.
[[AI]]:	"
Response = "	
	An area in the balcony [balcony] with (4, 2.5, 4). The microphone at (3.5, 0.5, 1.2), the voice at (2, 1.5, 1.6), the sound of footsteps [the sound of footsteps] at (0.5, 0.5, 1.2), the sound of loud talking [the sound of loud talking] at (2, 1.5, 1.6), the sound of vehicle noise [the sound of vehicle noise] at (2, 0.5, 2)."

Fig. 3: Examples of scene dynamic generation. Initially, the B (Background) component entails the user providing a clear and detailed description of the specified scene’s design and context (red font). Subsequently, the E (Examples) component requires the user to input few-shot prompts, which include a text description of the target scene (blue font). Following this, the generative chat model generates scene-based information that aligns with the predefined task background (green font). Finally, the T (Task) component involves the user specifying requirements for the dynamic generation of the scene (purple font).

A. DGSi: Dynamic Generation of Scene-based Information

Employing generative chat models structured within the BET prompt framework, the DGSi component facilitates the dynamic synthesis of tailored scene-based information for specific acoustic environments. An example of scene dynamic generation using the generative chat model BlueLM-Chat is provided in Fig. 3. To facilitate the production of specific information about scenes by the generative chat model, this paper introduces a new prompt framework, BET, which is an acronym for Background (B), Examples (E), and Task (T).

In summary, the E component of the BET effectively utilizes a structured query and response format to guide the generative chat model in producing detailed scene-based information. The Query can be represented as $\mathbf{Q}_n = [a_n, s_n]$, where n denotes the current example number, a is an adjective describing the scene (e.g., *Noisy* in Fig. 3), and s represents the type of the scene (e.g., *balcony*). The Response can be represented as $\mathbf{R}_n = [d_n, s_n, ml_n, sl_n, nt_n^1, nl_n^1, nt_n^2, nl_n^2, \dots, nt_n^m, nl_n^m]$, where m denotes the number of noise types included in the response, d refers to the scene dimensions (e.g., (4, 2.5, 4)), ml is the microphone location (e.g., (3.5, 0.5, 1.2)), sl is the speaker location (e.g., (2, 1.5, 1.6)), nt represents the noise type (e.g., *the sound of footsteps*), and nl is the noise location (e.g., (0.5, 0.5, 1.2)). The T can be represented as $\mathbf{T} = [a, s]$. Finally, the output from the generative chat model, which constitutes the dynamically generated scene-

based information, can be represented as scene-based information $\mathbf{SI} = [d, s, ml, sl, nt^1, nl^1, nt^2, nl^2, \dots, nt^m, nl^m]$.

B. SNAS: Scene-based Noise Addition for Speech

1) *Scene-based noise generation*: SNAS drawing inspiration from AudioLDM [42], the noise type $\mathbf{NT} = [nt^1, nt^2, \dots, nt^m]$ within the scene-based information can be utilized as prompt text to drive the TTA system for generating scene-based noise. Initially, $\mathbf{NT}$ is extracted from the DGSi module, and each nt^i as the prompt text $\mathbf{E}^y$ for the TTA system. Then LDM is employed to generate audio prior representations, facilitated by leveraging Contrastive Language-Audio Pretraining (CLAP) [46]. Utilize a Variational Auto-Encoder (VAE) [47] decoder to reconstruct the mel-spectrogram from the prior audio representations. Finally, HiFi-GAN [48] serves as a vocoder to convert the mel-spectrogram into a final audio output. This integrated process not only allows for the synthesis of scene-based noise that is coherent with the specified scene types but also significantly enhances the realism and immersion of the scene-based speech.

In the CLAP, audio samples are denoted as x and textual descriptions as y . It employs a text encoder $f_{\text{text}}(\cdot)$ and an audio encoder $f_{\text{audio}}(\cdot)$ to extract text embeddings $\mathbf{E}^y$ and audio embeddings $\mathbf{E}^x$ respectively. Once the CLAP model is trained, audio samples x are transformed into embeddings $\mathbf{E}^x$ within a space that is aligned with the text embeddings. The LDM starts with an initial distribution of Gaussian noise $p(z_N) \sim \mathcal{N}(0, I)$ and text embedding $\mathbf{E}^y$, and gradually generates audio prior representation z_0 through the following denoising process:

$$p_\theta(z_{0:N}|\mathbf{E}^y) = p(z_N) \prod_{t=n}^N p_\theta(z_{n-1}|z_n, \mathbf{E}^y) \quad (1)$$

$$p_\theta(z_{n-1}|z_n, \mathbf{E}^y) \sim \mathcal{N}(z_{n-1}; \mu_\theta(z_n, n, \mathbf{E}^y), \sigma_n^2 I) \quad (2)$$

Where p_θ is the model distribution, and is used to approximate the true data distribution. z_0 is the prior representation of the audio sample x within the space formed by the compressed representation of the mel-spectrogram $\mathbf{X} \in \mathbb{R}^{T \times F}$.

The mean and variance are parameterized as follows:

$$\mu_\theta(z_n, n, \mathbf{E}^y) = \frac{1}{\sqrt{\alpha_n}} \left(\left(z_n - \frac{\beta_n}{\sqrt{1 - \bar{\alpha}_n}} \right) \epsilon_\theta(z_n, n, \mathbf{E}^y) \right) \quad (3)$$

$$\sigma_n^2 = \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n} \beta_n \quad (4)$$

Where $\epsilon_\theta(z_n, n, \mathbf{E}^y)$ represents the predicted generative noise in the denoising process, α_n is a reparameterization of $1 - \beta_n$, and $\bar{\alpha}_n$ is defined as $\bar{\alpha}_n = \prod_{s=1}^n \alpha_s$. The sequence of β_n values follow a pre-defined schedule such that $0 < \beta_1 < \dots < \beta_n < \dots < \beta_N < 1$.

The VAE consists of an encoder and a decoder with stacked convolutional modules. The encoder compresses the mel-spectrogram $\mathbf{X}$ into a latent space $z \in \mathbb{R}^{C \times \frac{T}{r} \times \frac{F}{r}}$, where r

denotes the compression ratio. The decoder aims to reconstruct the mel-spectrogram $\hat{\mathbf{X}}$ from the audio prior representation $\hat{z}_0$ generated by the LDM. For the vocoder, HiFi-GAN is used from the mel-spectrogram $\hat{\mathbf{X}}$ to generate the scene-based noise samples $\hat{\mathbf{A}}$.

2) *Scene-based noise addition*: Drawing inspiration from pyroomacoustics [21], we can leverage the scene-based information to model the scene as a shoebox-shaped room. From the scene-based information obtained in the DGS module, extract the scene dimensions d and the various locations, such as microphone location ml , speaker location sl , and noise location nl . Combine the extracted scene parameters with the scene-based noise $\mathbf{SN} = [\hat{A}^1, \hat{A}^2, \dots, \hat{A}^m]$ to serve as input for the RIR filter. This means the simulated room dimensions are equal to d , the microphone location $l = ml$, and the real sound source s_0 can be either original speech with sl or $s_0 = [\hat{A}^i, nl^i]$. First, define a room with some sound sources and a microphone to create the simulated scene. The actual audio is added as the original audio sample to the sound sources. Then, utilize the ISM to find all mirror sources in a specified order. Once the locations of the mirror sources and their visibility from the microphone location are determined, use this information to construct the RIR. The RIR between the microphone at location l and a real sound source s_0 at location sl , considering a set of visible mirror sources $V_l(s_0)$, can be calculated as:

$$a_r(s_0, n) = \sum_{s \in V_l(s_0)} \frac{(1 - \alpha)^{R(s)}}{4\pi \|l - s\|} \delta_{LP} \left(n - F_s \frac{\|l - s\|}{c} \right) \quad (5)$$

Where $R(s)$ determines the reflection order of the sound source s . $\alpha \in [0, 1]$ represents the absorption coefficient of the walls. c is the speed of sound in the air. δ_{LP} is the windowed sinc function. n represents the current time step in the simulation. F_s is the sampling rate, indicating how many time steps are in one second.

$$\delta_{LP}(t) = \begin{cases} \frac{1}{2} \left(1 + \cos\left(\frac{2\pi t}{T_w}\right) \right), & -\frac{T_w}{2} \leq t \leq \frac{T_w}{2} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Where T_w is the width of the window used in the windowed sinc function. The width of the window determines the trade-off between computational efficiency and the accuracy of the windowed sinc function approximation. Finally, by convolving the audio signal with RIR, obtain the scene-based speech.

IV. EXPERIMENTAL SETTINGS

This section is divided into five components, including benchmark datasets, baseline models, generative chat models, evaluation metrics, and experimental parameters.

A. Benchmark datasets

As discussed in Section I, DGSNA is proposed to address the limited robustness of speech systems when transitioning from controlled laboratory settings to real-world applications. To demonstrate effectiveness of DGSNA for both ASR and

KWS tasks, we designed experiments using a clean dataset for training (simulating laboratory conditions) and a noisy dataset for testing (simulating real-world applications).

For the ASR experiment, we used Aishell-1 [49] as the training set and WenetSpeech [50] as the test set.

For the KWS domain, we were unable to locate datasets analogous to Aishell-1 and WenetSpeech (i.e., a clean training set and a noisy test set). Therefore, we used the HeySnips [51] dataset, which contains background noise in both training and testing sets, for our KWS experiments.

- **Aishell-1**: Aishell-1, released by Beijing Shell Shell Technology Co., Ltd., comprises over 170 hours of Mandarin speech data from 400 speakers. It is a subset of the 500-hour multi-channel AISHELL-ASR0009 corpus, designed for various speech and speaker processing tasks.
- **WenetSpeech**: WenetSpeech is a multi-domain Mandarin corpus comprising over 22,400 hours of speech, including 10,000+ hours of high-quality labeled speech, 2,400+ hours of weakly labeled speech, and approximately 10,000 hours of unlabeled speech. WenetSpeech comprises three subsets: (a) Dev, is specifically designed dataset for some speech tools that require cross-validation in training. (b) Test_Net, is a matched test set from the internet. (c) Test_Meeting, is a mismatched test set since it is a far-field, conversational, spontaneous, and meeting speech dataset.
- **HeySnips**: The keyword ‘‘Hey Snips’’ (pronounced without a pause) was used. The dataset includes diverse English accents and recording environments, comprising approximately 11,000 wake-word utterances and 86,500 (96 hours) negative examples.

During data preprocessing, DGSNA was applied probabilistically ; in other words, noise augmentation was not performed on every speech sample. The probability of adding noise in this process is referred to as the Add Noise Rate (ANR).

B. Baseline models

We selected Conformer [52] as the baseline for ASR experiments, leveraging the WeNet [53] framework. For KWS experiments, we utilized the Multi-scale Depthwise Temporal Convolution (MDTC) [54] model as the baseline, implemented within the WeKws [55] framework.

- **Conformer**: Conformer adds a Convolution module based on transformer [56] so it can capture both local and global context and get better results on different ASR tasks.
- **WeNet**: WeNet implements U2, a novel two-pass approach that unifies streaming and non-streaming end-to-end (E2E) speech recognition within a single model. U2 consists of three parts, Shared Encoder, CTC Decoder and Attention Decoder. We use Conformer as our shared encoder, while the Attention Decoder is transformer-based.
- **MDTC**: MDTC explicitly fuses multi-scale features from different hidden layers with different receptive fields and models long-range temporal features with efficient dilated depthwise temporal convolution.



Fig. 4: Overview of DGSNA data generation.

- **WeKws:** WeKws consists of four parts, starting with a global Cepstral Mean and Variance Normalization (CMVN) layer to normalize the input acoustic features to a normal distribution. This is followed by a linear layer which maps the dimensions of the input features to the required dimensions. Then there is a backbone network. At the end of the model, there are several binary classifiers. We used MDTC as the backbone of our model.

C. Generative chat models

This paper references C-Eval [57], an evaluation benchmark specifically designed to assess the capabilities of generative chat models in both Chinese and English contexts. In consideration of the computational resources typically available to most researchers, the selection of models was strategically based on their performance in the C-Eval rankings. Consequently, the experiment included the following generative chat models: BlueLM-7B-Chat [58], Yi-6B-Chat [59], ChatGLM3-6B [60], and Qwen-7B-Chat [32].

D. Evaluation metrics

When evaluating the performance of our speech processing systems, we utilize distinct metrics tailored to the objectives of each individual task.

For ASR experiments, we employ the Word Error Rate (WER), which provides a comprehensive measure of transcription accuracy by quantifying the number of substitutions, insertions, and deletions required to align the system’s output with the reference transcript. A lower WER signifies better ASR performance.

In contrast, for KWS experiments, our focus shifts to detection accuracy, where we use the False Rejection Rate (FRR) and the False Alarm Per Hour (FA / Hour). The FRR indicates the percentage of times the system fails to detect a spoken keyword when it is present, while the FA / Hour measures how frequently the system incorrectly identifies a keyword in the absence of one. These KWS metrics provide a clear understanding of the system’s ability to reliably detect target keywords while minimizing erroneous detections.

E. Experimental parameters

Additionally, within the TTA system employed in this experiment, the chosen model was AudioLDM-I-full [42]. The number of diffusion steps (*ddim_steps*) for the Denoising Diffusion Implicit Model (DDIM) [61] was set to 200, controlling the iterations of the denoising process. A guidance scale (*guidance_scale*) of 2.5 was employed to steer the model’s output based on provided guidance, enhancing alignment with target

audio characteristics. The generated audio duration (*duration*) was set to 5.0 seconds.

In our experiment, pyroomacoustics was selected as the library to implement the RIR filter. The reverberation time (*rt60*) is set to 0.5 seconds, representing the time required for the sound pressure level to decrease by 60 decibels (*dB*). The sampling rate (*fs*) is configured to 16,000 Hz. The maximum specified order (*max_order*) is limited to 1. The reasons for these choices are as follows:

In indoor scenes, where the physical space is constrained and sound sources are fewer, this setting allows for the simulation of first-order echoes reflecting from the nearest surfaces without overly complicating the acoustic model. This method effectively models essential reflections while preventing the unrealistic accumulation of reverberation that might result from higher-order reflections. Conversely, in outdoor scenes, echoes and reverberations are typically negligible due to the absence of confining structures. In these scenarios, reflections, due to their minimal impact, can conceptually be treated as additional sound sources, simplifying the acoustic modeling process and focusing on direct sound transmission. This analytical approach offers the advantage of simulating extensive acoustic environments without requiring intricate modeling of complex interactions between sound waves and environmental features, particularly beneficial in large, open spaces where such interactions are less defined.

V. EXPERIMENTAL RESULTS

A. DGSNA vs. other noise addition methods

The process of data generation using DGSNA, simplified in Fig. 4, highlights that its key difference from other noise addition methods lies in the degree of domain randomization during Step 2. To further compare DGSNA with these methods, we conducted comparative experiments in both the ASR and KWS domains, respectively, and the results of the experiments are shown in Table I. and Fig. 5, respectively.

The experimental results suggest that DGSNA’s effects are similar to those of SpecAugment in practical applications. However, we observed that combining DGSNA and SpecAugment yields even better results. We believe DGSNA adds acoustic environments to clean speech, enabling the model to learn robustness against scene-based noise. Simultaneously, SpecAugment masks portions of the spectrum, improving the model’s feature extraction capabilities. These methods are complementary, and to improved model generalizability, we recommend using multiple noise addition methods in combination.

TABLE I
RESULTS OF THE ASR EXPERIMENT. ALL MODELS SHARE THE SAME ARCHITECTURE

Model	Train Set	Stream Decode	Test WER(%) ↓			
			Aishell-1		WenetSpeech	
			Test	Dev	Test_Net	Test_Meeting
gaussian noise	Aishell-1	True	6.22	33.07	51.31	54.16
		False	5.81	32.51	50.75	53.85
SpecAug	Aishell-1	True	5.33	29.90	48.94	53.99
		False	4.93	29.16	48.29	53.60
DGSNA	Aishell-1	True	6.01	32.68	50.70	50.69
		False	5.44	32.00	50.03	50.35
DGSNA & SpecAug	Aishell-1	True	5.59	30.01	47.18	49.10
		False	5.00	29.20	46.43	48.61

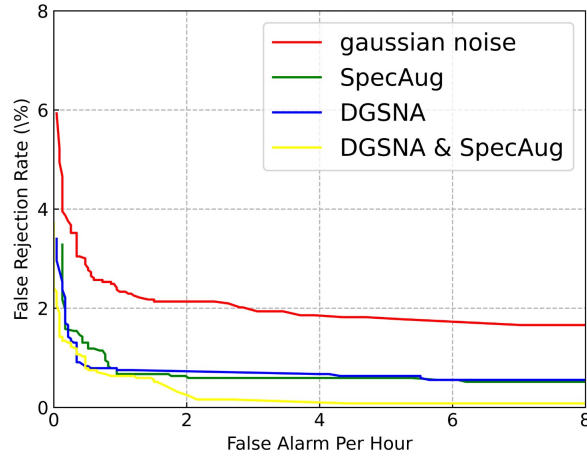


Fig. 5: Results of the KWS experiment.

B. Comparison of different ANR

Table II. and Table III. presents the ASR experimental results. Introducing DGSNA during training significantly improves performance in both known noisy scenarios (Aishell-1 with $ANR > 0$) and, to a lesser extent, unknown scenarios (WenetSpeech), achieving up to 11.21% relative improvement. This improvement comes at the cost of slight performance degradation in clean scenarios (Aishell-1 with $ANR = 0$). The DGSNA application rate (ANR) significantly influences results. We hypothesize that a low ANR (10%) results in insufficient data augmentation, while a high ANR (40%) leads to overfitting. An ANR of 20% proved optimal.

Fig. 6. presents the KWS experimental results. Introducing DGSNA during training significantly improves performance in known noisy scenarios, with an ANR of 20% once again demonstrating optimal effectiveness.

C. Comparison of different generative chat models

1) *BET prompt*: The effectiveness of a generative chat model in dynamic scene simulation significantly hinges on the adaptability of the BET prompts to the model's specific input-output structure.

To ensure compatibility across diverse model architectures, tailoring the BET prompt to the operational mode of the target generative chat model is crucial. Generative chat models are commonly classified into two types based on their parameter-passing methods: (a) Single Parameter Input Models: In this configuration, models such as BlueLM-Chat concatenate the current chat input and historical chat information into a single consolidated parameter, typically termed the prompt. (b) Dual Parameter Input Models: Conversely, models like ChatGLM3 are designed to process the current chat input and historical chat information as two separate parameters, referred to as the prompt and history. As illustrated in Fig. 7, we developed tailored prompts for two distinct models, BlueLM-Chat and ChatGLM3, corresponding to their respective single and dual parameter input frameworks.

For single parameter input models like BlueLM-Chat, where the prompt requires the integration of the current chat input with historical chat information into a single parameter, the construction of the BET prompt is sequential and cumulative. Each chat interaction builds upon the previous one. The BET prompt is constructed in the order of Background, Examples, and Task. For each Example, the prompt is further refined by concatenating query and response.

TABLE II
RESULTS OF THE ASR EXPERIMENT (AISHELL-1). ALL MODELS SHARE THE SAME ARCHITECTURE AS THE BASELINE, WITH SUBSCRIPT NUMBERS INDICATING THE RELATIVE IMPROVEMENT

Model	Train Set ANR	Stream Decode	Test WER(%) ↓				
			Aishell-1 ANR				
			0%	10%	20%	30%	40%
Baseline	0%	True	5.33	9.90	14.30	19.04	23.72
		False	4.93	9.47	13.86	18.51	23.23
DGSNA_10	10%	True	5.52 ^{+3.56%}	6.87 ^{-30.60%}	8.55 ^{-40.20%}	9.52 ^{-50.00%}	11.08 ^{-53.28%}
		False	4.98 ^{+1.01%}	6.27 ^{-33.79%}	7.90 ^{-43.00%}	8.82 ^{-52.35%}	10.36 ^{-55.40%}
DGSNA_20	20%	True	5.59 ^{+4.87%}	6.77 ^{-31.61%}	7.91 ^{-44.68%}	9.03 ^{-52.57%}	10.11 ^{-57.37%}
		False	5.00 ^{+1.41%}	6.13 ^{-35.26%}	7.22 ^{-47.90%}	8.33 ^{-54.99%}	9.30 ^{-59.96%}
DGSNA_30	30%	True	5.75 ^{+7.87%}	6.77 ^{-31.61%}	7.69 ^{-46.22%}	8.81 ^{-53.72%}	9.67 ^{-59.23%}
		False	5.17 ^{+4.86%}	6.13 ^{-35.26%}	6.97 ^{-49.71%}	8.05 ^{-56.50%}	8.91 ^{-61.64%}
DGSNA_40	40%	True	6.06 ^{+13.69%}	6.97 ^{-29.59%}	8.15 ^{-43.00%}	9.23 ^{-51.52%}	10.19 ^{-57.04%}
		False	5.40 ^{+9.53%}	6.26 ^{-33.89%}	7.33 ^{-47.11%}	8.41 ^{-54.56%}	9.32 ^{-59.87%}

TABLE III
RESULTS OF THE ASR EXPERIMENT (WENETSPEECH). ALL MODELS SHARE THE SAME ARCHITECTURE AS THE BASELINE, WITH SUBSCRIPT NUMBERS INDICATING THE RELATIVE IMPROVEMENT

Model	Train Set ANR	Stream Decode	Test WER(%) ↓			
			Aishell-1 ANR	WenetSpeech		
			0%	Dev	Test_Net	Test_Meeting
Baseline	0%	True	5.33	29.90	48.94	53.99
		False	4.93	29.16	48.29	53.60
DGSNA_10	10%	True	5.52 ^{+3.56%}	29.89 ^{-0.03%}	47.50 ^{-2.94%}	49.87 ^{-7.63%}
		False	4.98 ^{+1.01%}	29.13 ^{-0.10%}	46.77 ^{-3.14%}	49.42 ^{-7.79%}
DGSNA_20	20%	True	5.59 ^{+4.87%}	30.01 ^{+0.36%}	47.18 ^{-3.59%}	49.10 ^{-9.05%}
		False	5.00 ^{+1.41%}	29.20 ^{+0.13%}	46.43 ^{-3.85%}	48.61 ^{-9.30%}
DGSNA_30	30%	True	5.75 ^{+7.87%}	30.33 ^{+1.43%}	47.94 ^{-2.04%}	48.02 ^{-11.05%}
		False	5.17 ^{+4.86%}	29.48 ^{+1.09%}	47.18 ^{-2.29%}	47.59 ^{-11.21%}
DGSNA_40	40%	True	6.06 ^{+13.69%}	31.52 ^{+5.41%}	50.09 ^{+2.34%}	50.43 ^{-6.59%}
		False	5.40 ^{+9.53%}	30.64 ^{+5.07%}	49.35 ^{+2.19%}	49.97 ^{-6.77%}

For dual parameter input models like ChatGLM3, where the current chat input and the historical chat information are fed as separate parameters. Specifically, history is formatted as a sequence that includes the background, followed by alternating queries and responses from the examples. In contrast to the history, the prompt is straightforward and focused, being comprised solely of the Task. In conclusion, the construction of a BET prompt, as described, can be finely adjusted based on the specific input mode of various generative chat models.

2) *Filter metrics*: To ensure that the generated scenes are suitable for subsequent scene-based noise addition, our research group developed and implemented a set of filter metrics. These metrics are specifically designed to assess the quality and applicability of responses generated by different generative chat models, effectively screening out those that do not meet established criteria. Table IV. presents the performance of four generative chat models against these filter metrics, and Fig. 8. provides illustrative examples for each metric, with red font denoting incorrect content. The specifics of these filter metrics are:

TABLE IV
PERFORMANCE OF FOUR GENERATIVE CHAT MODELS AGAINST FILTER METRICS

Model	Metric (%) ↓			
	#1	#2	#3	#4
BlueLM-7B-Chat	37.7	7.1	25.0	10.7
Yi-6B-Chat	2.2	2.2	56.8	0
ChatGLM3-6B	43.3	15.7	3.9	0
Qwen-7B-Chat	2.2	6.8	12.5	0

- **Response Error**: Verifies adherence to the 3-shot example format and the presence of essential information, such as scene dimensions and coordinates.
- **Microphone Overlapping Source**: Ensures the microphone's position does not overlap any sound sources, preventing one sound from completely masking others.
- **Location Exceed Dimensions**: Checks if locations in the response are within the defined scene dimensions.

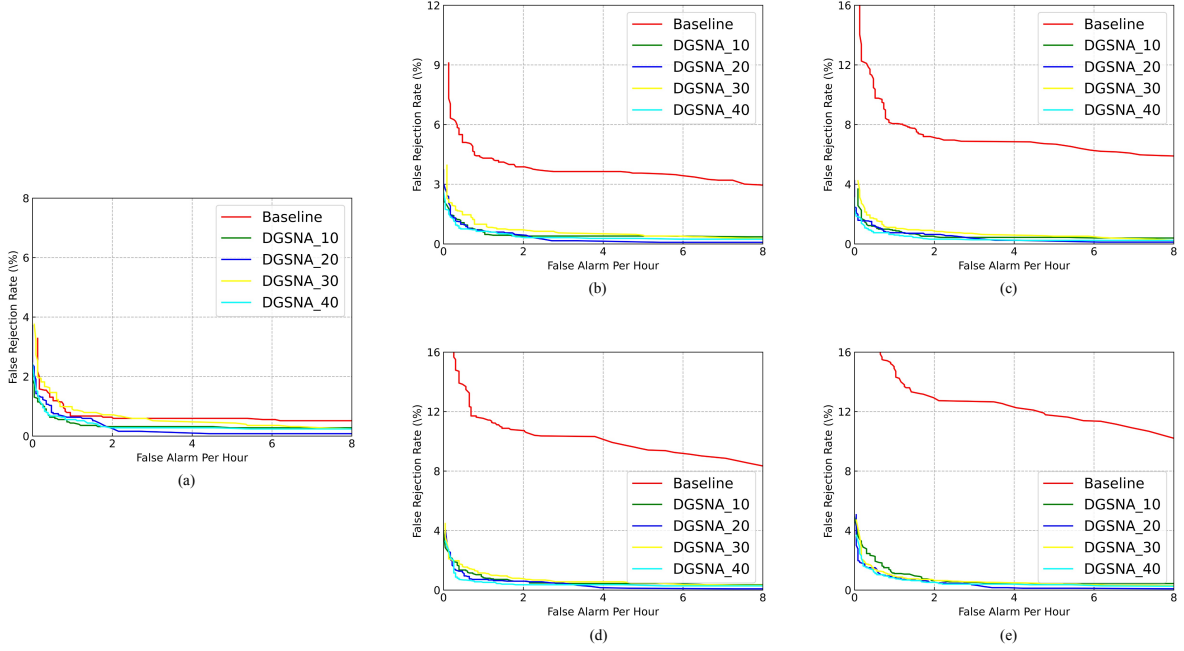


Fig. 6: Results of the KWS experiment. All models share the same architecture as the baseline. The five subplots, presented alphabetically, show performance on the test set at ANRs of 0%, 10%, 20%, 30%, and 40%.

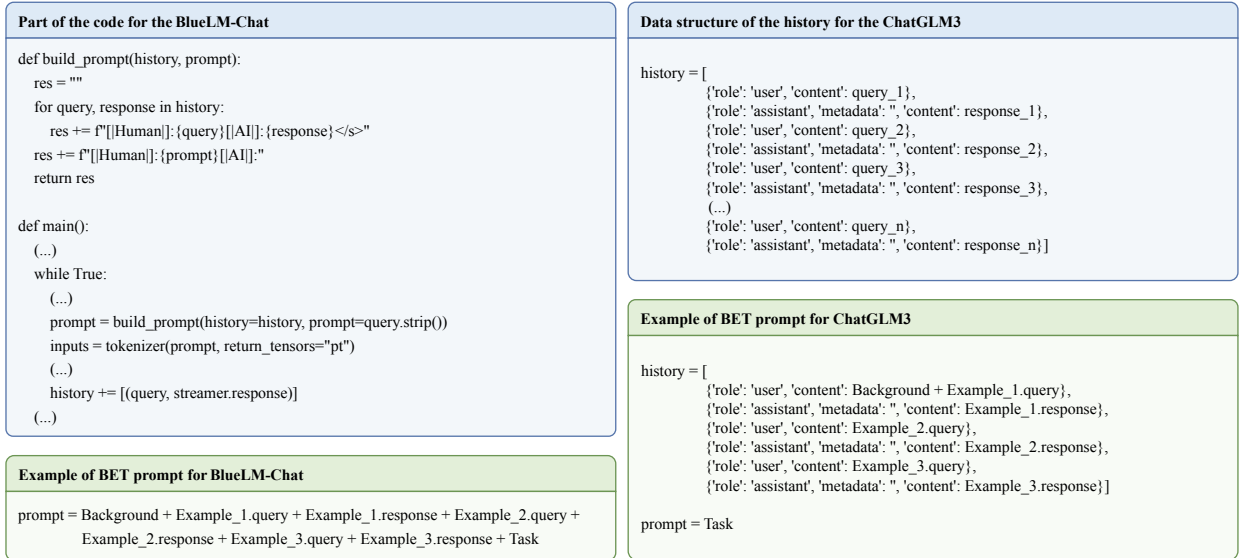


Fig. 7: Construction of the BET prompt.

- **Types Less Than Target:** Measures whether the number of noise types specified in the response meets or exceeds a predefined target, set to 2 for this experiment.

D. Example analysis

By leveraging the BET prompt framework alongside generative chat models, TTA systems, and RIR filters, we can effectively implement the DGSNA. This section elucidates the specific process through which scene-based speech is generated, illustrated by the analysis of examples.

1) *Correct Example:* This subsection provides a detailed analysis of the workflow for generating scene-based speech,

using the sample depicted in Fig. 9 as an illustration. This process underscores the comprehensive steps undertaken to ensure the production of high-quality, contextually appropriate scene-based speech utilizing DGSNA.

- **Original Prompt:** The process begins with the input text “noisy pedestrian street.”
- **BET Prompt:** The BET prompt has been strategically adapted to conform to the input-output structure and historical context management strategy employed by the Qwen-7B-Chat model.
- **Scene-based Information:** The adapted BET prompt is fed into the Qwen-7B-Chat model, which processes

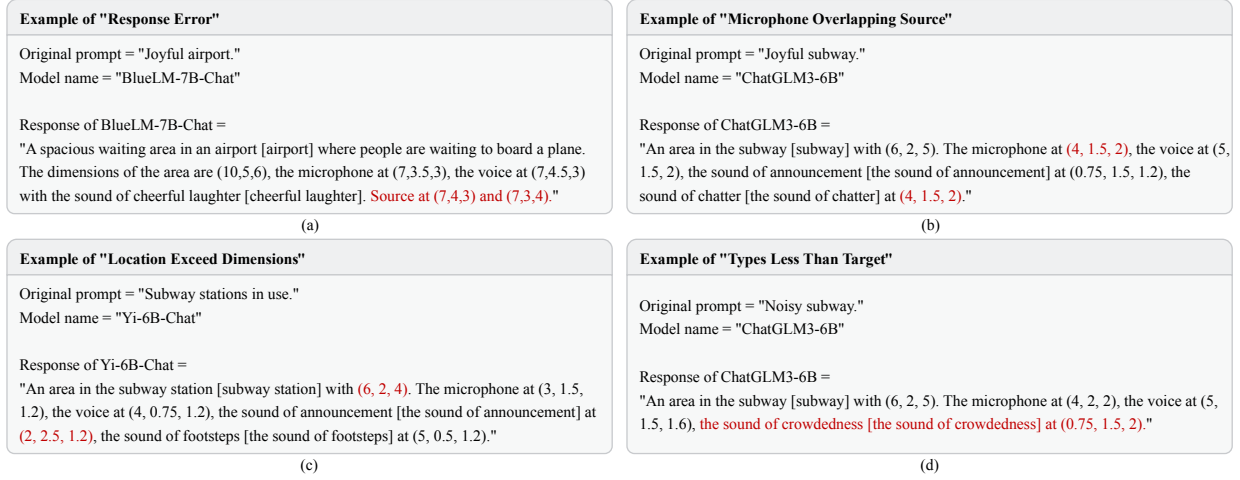


Fig. 8: Examples of each filter metric.

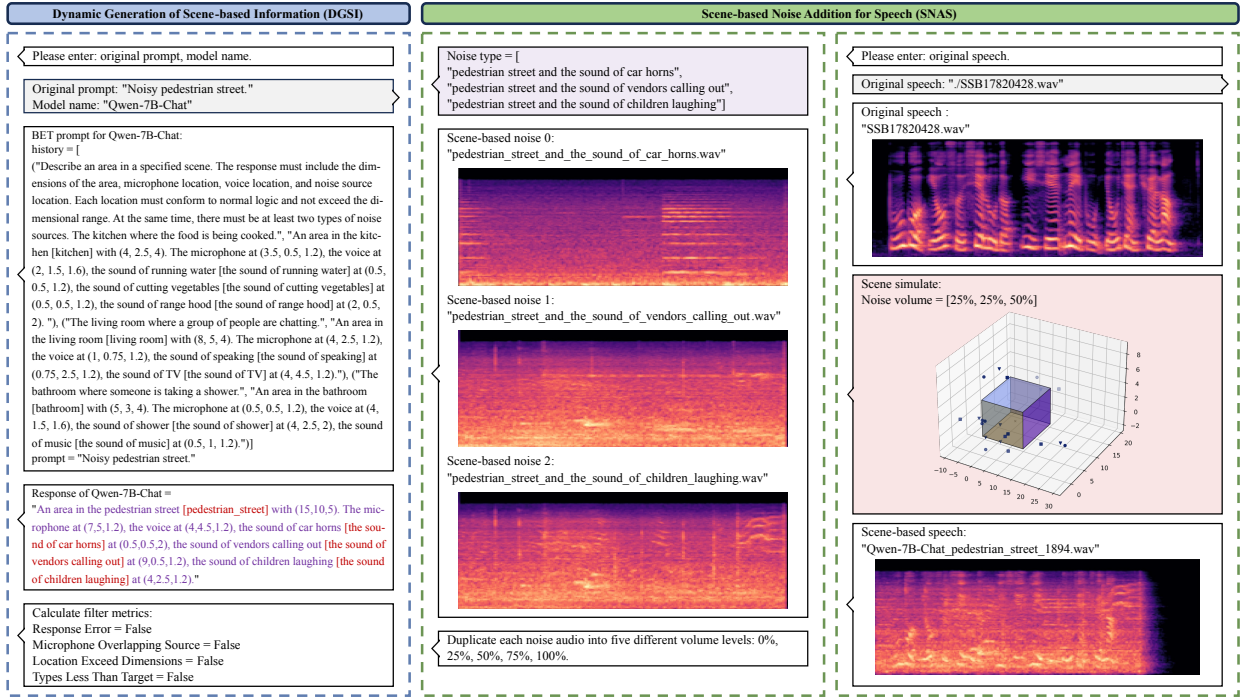


Fig. 9: Workflow for generating scene-based speech.

the information and dynamically generates scene-based information.

- **Filter Metrics:** The generated scene-based information is assessed using predefined filter metrics to confirm its suitability for subsequent processing.
- **TTA System:** The TTA system utilizes each noise type specified in the scene-based information as a text embedding. These embeddings undergo a denoising procedure to generate precise audio prior representations. Subsequently, these representations are converted into actual audio samples through the combined use of a decoder and a vocoder.
- **Scene-based Noise:** Each audio sample is generated at five distinct volume levels, mimicking real-world variations in sound intensity. To improve the realism of

scene-based noise addition and more accurately simulate real-world acoustic environments, our research group incorporated a parameter for noise source volume levels. Employing torchaudio [62], five distinct volume levels—0%, 25%, 50%, 75%, and 100%—were generated for each noise audio sample. During simulation, a random selection process determined the active volume level for each noise source.

- **Original Speech:** The original speech is obtained.
- **Scene-based Speech:** The RIR filter uses the scene dimensions and source coordinates provided in the scene-based information to construct the RIR. Subsequently, it convolves both the generated scene-based noise and original speech with the corresponding RIR, culminating in the production of the scene-based speech.



Fig. 10: Workflow for generating four specific examples.

We present the scene visualization and the example speech's spectrogram in Fig. 9. Squares, inverted triangles, and circles each denote a unique type of sound source and their mirror sources. The number of mirror sources (e.g., 18) is calculated based on the product of the number of original sound sources (e.g., 3), the number of scene surfaces (e.g., 6), and the maximum specified order of reflection (e.g., 1). In outdoor scenes, these mirror sources are treated as additional sound sources of the same type. By including both original and mirror sources, the total number of sound sources in the scene amounts to 21. This comprehensive inclusion ensures a more detailed and accurate simulation, as it accounts for multiple interactions and the resultant acoustic effects within the scene.

2) *Incorrect Example*: In this section, the analysis of four specific examples from Section V-C is detailed, each illustrating a distinct issue encountered in the application of the DGSNA. Each example is associated with a particular filter metric that has been triggered due to identified errors, demonstrating how the system can prevent incorrect or inadequate data from progressing through the audio generation process. Fig. 10. outlines the general workflow for four specific examples. Here is the analysis of specific issues:

- **First Example (Fig. 10.a):** The response fails to adhere to the 3-shot format provided, contains logical inconsistencies, lacks essential details such as scene dimensions and coordinate points, and fails to provide translations.
- **Second Example (Fig. 10.b):** The position of the microphone overlap the source of the crowd chatter, which will result in avoiding the crowd chatter from completely

masking the other sounds.

- **Third Example (Fig. 10.c):** Locations mentioned in the response exceeds the defined dimensions of the scene. Such discrepancies can lead to computational errors (e.g., ValueError) when the data is used in subsequent processes like the RIR filter.
- **Fourth Example (Fig. 10.d):** The response includes fewer noise types than the targeted number set in the BET prompt, which is specified as 2. Insufficient variety in noise types compromises the depth and authenticity of the scene simulation, reducing its effectiveness in creating a believable acoustic environment.

VI. CONCLUSION

This paper introduces DGSNA, a novel generative scene-based noise addition method. DGSNA implements DGSI by combining the BET prompt framework with a generative chat model, while SNAS is implemented through the deployment of a TTA system and RIR filters. This methodology effectively generates scene-based information and noise dynamically, thereby improving scene-based noise addition for speech and addressing the reliance of existing noise addition methods on preexisting scene-based noise and information. The methodology and efficacy of DGSNA were rigorously evaluated through several experiments. Firstly, comparative experiments with current mainstream noise addition methods demonstrated that integrating DGSNA with other methods yielded superior results. Secondly, DGSNA was employed as a noise addition method for both ASR and KWS, exploring the systematic enhancement effects of varying ANR,

achieving relative improvements of up to 11.2%. Finally, the generalizability of DGSNA across diverse generative dialogue models was explored, showcasing the inherent robustness of DGSNA. Future work will explore applying DGSNA to additional speech tasks, training models across a wider range of simulated acoustic environments to enhance their robustness and real-world performance.

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